Laboratory of Biomedical Optics – Biophotonics

Exercise 1: Time-Resolved Photon Migration in an Infinite Medium

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1 Introduction

In the near-infrared (NIR) spectral window (650–900 nm), biological tissue behaves as a strongly scattering and weakly absorbing turbid medium ($\mu_s' \gg \mu_a$). Light transport can be approximated by the **Diffusion Equation (DE)** for the optical fluence rate $\Phi(\mathbf{r}, t)$ [photons/(cm$^2 \cdot$ s)]:

$$\frac{1}{v}\frac{\partial \Phi(\mathbf{r}, t)}{\partial t} - D\nabla^2\Phi(\mathbf{r}, t) + \mu_a\Phi(\mathbf{r}, t) = S(\mathbf{r}, t) \tag{1}$$

Green's Function in an Infinite Medium

For an instantaneous point pulse at position $\mathbf{r}_0$ and time $t = 0$ ($S(\mathbf{r}, t) = \delta(\mathbf{r} - \mathbf{r}_0)\delta(t)$) in an infinite homogeneous medium, the analytical solution is:

$$\Phi(r, t) = v\,(4\pi Dvt)^{-3/2} \exp\left(-\frac{r^2}{4Dvt}\right) \exp(-\mu_a vt) \tag{2}$$

where $r = \|\mathbf{r} - \mathbf{r}_0\|$ is the source-detector distance.
The three factors represent:

1. $(4\pi Dvt)^{-3/2}$: 3D geometric expansion of the diffusing photon cloud over time.
2. $\exp\left(-\frac{r^2}{4Dvt}\right)$: Gaussian spatial dispersion centered at the source.
3. $\exp(-\mu_a vt)$: Exponential attenuation (Beer-Lambert law) along the pathlength $s = vt$ traversed by photons detected at time t.

Physical Parameters:

- **Refractive index** n: Typically $n \approx 1.4$ in soft tissue.
- **Speed of light in tissue** v: $v = c_0/n$, with $c_0 = 0.03\,\text{cm/ps}$ (3×10^8 m/s).
- **Absorption coefficient** μ_a: [cm^{-1}], probability of absorption per unit length.
- **Reduced scattering coefficient** μ_s': [cm^{-1}], $\mu_s' = \mu_s(1 - g)$, where $g \approx 0.9$ is the anisotropy factor.
- **Diffusion coefficient** D: In the diffusion regime ($\mu_s' \gg \mu_a$), $D \approx \frac{1}{3\mu_s'}$.

2 Objectives

1. Implement the analytical Green's function in a MATLAB function `TDInfinite.m`.
2. Simulate the time-resolved Distribution of Time of Flight (DTOF).
3. Systematically study the effect of absorption (μ_a), scattering (μ_s'), and distance (r).
4. Visualize the time evolution of the 3D photon distribution on a 2D cross-section.

3 Step-by-Step Instructions

Step 1: Implement the Green's Function (`TDInfinite.m`)

Create the function `TDInfinite.m`. Implement the Green's function following the comments in the code to correctly manage the parameters.

Step 2: Simulation Setup and Baseline Curve

In the script `Exercise1_Infinite.m`, define the baseline parameters and plot the first DTOF curve.

```
n = 1.4;
dt = 5;                    % time step [ps]
N = 2000;                  % number of points
t = (0:N-1) * dt + dt/2;  % time vector [ps]
mua = 0.01;                % [cm^-1]
musp = 10;                 % [cm^-1]
SourcePos = [0, 0, 0];     % [cm]
DetPos = [2, 0, 0];        % [cm] (r = 2 cm)
```

Step 3: Parametric Studies

Influence of Absorption (μ_a)

- Fix $r = 2\,\text{cm}$ and $\mu_s' = 10\,\text{cm}^{-1}$.
- Vary $\mu_a \in [0,\, 0.2]\,\text{cm}^{-1}$ with step $0.01\,\text{cm}^{-1}$.
- Plot the curves on a semilogarithmic scale (`semilogy`).

How does μ_a affect the late-time tail of the curve?

Influence of Reduced Scattering (μ_s')

- Fix $r = 2\,\text{cm}$ and $\mu_a = 0.01\,\text{cm}^{-1}$.
- Vary $\mu_s' \in [5,\, 20]\,\text{cm}^{-1}$ with unit increments.
- Plot the curves on a semilogarithmic scale.

How does μ_s' affect the peak arrival time ($t_{\max}$) and curve broadening? Why do the tails have identical slopes?

Influence of Source-Detector Distance (r)

- Fix $\mu_a = 0.01\,\text{cm}^{-1}$ and $\mu_s' = 10\,\text{cm}^{-1}$.
- Vary the detector position along x: $x_{\text{det}} \in [0.5,\, 4.0]\,\text{cm}$ with step $0.5\,\text{cm}$.
- Plot the curves on a semilogarithmic scale.

How do peak position and peak intensity scale with increasing distance?

Step 4: 3D Spatial Fluence Evolution (Animation)

Compute the spatial distribution of fluence on a 2D slice (xz-plane at $y = 0$) as time evolves. Use the function `drawnow`.

4 Extra questions

1. Calculate the derivative $\frac{\partial \Phi}{\partial t} = 0$ to find the analytical peak arrival time $t_{\max}$ as a function of r, D, v, μ_a.

2. Explain why time-domain measurements allow simultaneous, decoupled estimation of both μ_a and μ_s', unlike continuous-wave (CW) measurements at a single distance.